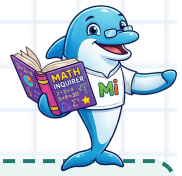
FILL THE GRID!

Name _____ Date _____

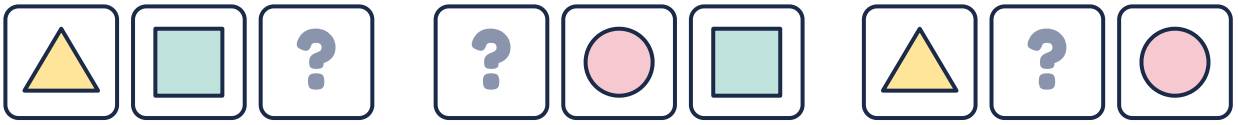


After question 2 you may choose: carry on, or try **Think harder**.

Use your tiles on mat 1 to check. Look along the row and down the column.

No.S1

Each row has a circle, a square and a triangle. Which shape goes in the box with the ? mark in rows A, B and C?



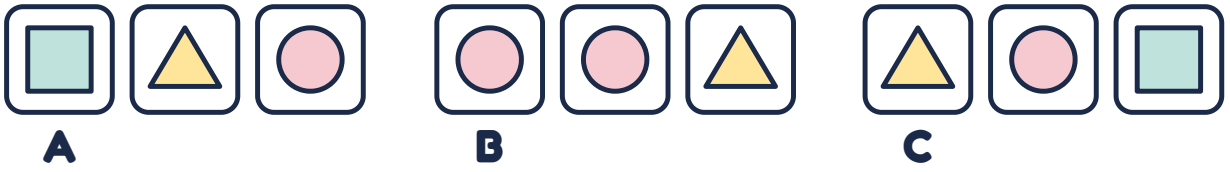
(a) Row A. Circle one. **circle** **square** **triangle**

(b) Row B. Circle one. **circle** **square** **triangle**

(c) Row C. Circle one. **circle** **square** **triangle**

No.S2

Look at the three rows. Tick True or False for each statement.



Statement	True	False
Row A has one of each.	☐	☐
Row B has one of each.	☐	☐
Row C has one of each.	☐	☐

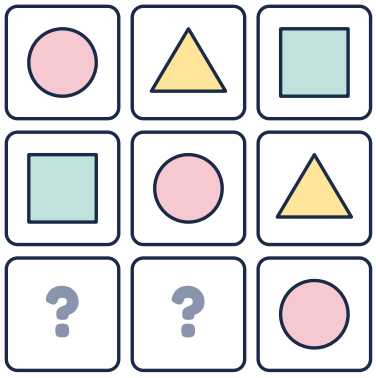
Put a tick (✓) in the correct box.

Name _____ Date _____

Check the rows. Then check the columns.

No.S3

Every row and every column has a circle, a square and a triangle. Look at the bottom left box.

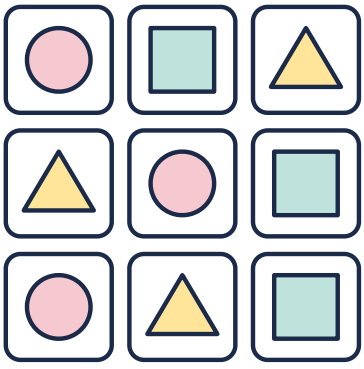


(a) Which shape goes in the bottom left box? Circle one. **circle** **square** **triangle**

(b) Which reason tells us? Circle one. **The column has a circle and a square.** **The row has a circle and a square.**

No.S4

A child fills the grid in the picture. Every row has one of each. Is the grid correct?



Circle Yes or No. **Yes** **No**

No.S5

Every row of a 3 by 3 grid has one circle, and every column has one circle. How many circles are in the whole grid?

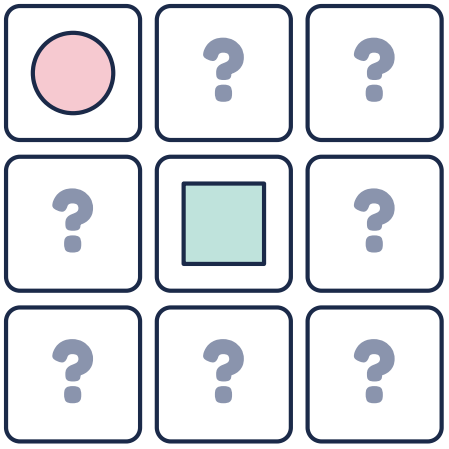
There are _____ circles in the grid.

Name _____ Date _____

Use your tiles or draw it. Choose any question, in any order.

No.T1

Mi has a grid with only two shapes in it. Every row and every column has one of each. Which shape goes in each box?



(a) Top middle box. **circle** **square** **triangle**

(b) Middle right box. **circle** **square** **triangle**

(c) Bottom right box. **circle** **square** **triangle**

No.T2

Every row and every column of a grid has a circle, a square and a triangle. How many different shapes could go in each box?

(a) Circle in its row, circle in its column: ☐ shapes.

(b) Triangle in its row, square in its column: ☐ shape.

No.T3

A grid is correct: every row and every column has one of each. A child changes it. Test each change with your tiles. Tick True or False.

Statement	True	False
Swap the top row and the bottom row. The new grid is still correct.	☐	☐
Swap two boxes in the middle row. The new grid is still correct.	☐	☐

Put a tick (✓) in the correct box.

Name _____ Date _____

Choose any question. You do not have to finish them all.

No.G1

Four by Four. Every row and every column has the digits 1, 2, 3 and 4 once each. Write the digits in the empty boxes.

1		2	3
3		4	2
	3		4
	2		

No.G2

Four by Four, with fewer clues. Only six digits are given. Every row and every column has 1, 2, 3 and 4 once each. Which digit goes in each box marked a, b and c? The other empty boxes are not asked.

2	1		a
	4		
b	3	1	
	c	3	

Box a is _____. Box b is _____. Box c is _____.

No.G3

A 5 by 5 grid uses the digits 1, 2, 3, 4 and 5. Every row and every column has each digit once.

(a) How many boxes are in the grid? _____

(b) How many boxes hold a 4 or a 5? _____

Name _____ Date _____

Investigation: Grid builder

First: Row 1 is 1 3 2. Row 2 is 3 2 1. Write row 3: _____ _____ _____

Next: Put a 2 in the middle of a 3 by 3 grid. Every row and column has 1, 2 and 3 once. Write each grid you can make on its own line, row by row. The first one is 132 321 213. A turned grid is a different grid. How many are there?

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

A row has a circle and a square. Which shape is missing from the row? Circle one.

circle **square** **triangle**

No.E2

Which row has one of each? Choose the correct answer. Write its number in the bracket.

(1) circle, circle, square

(2) square, triangle, circle

(3) triangle, triangle, circle

(4) circle, square, square

( )

No.E3

A box has a circle in its row and a square in its column. Which shape goes in the box? Circle one.

circle **square** **triangle**

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well: _____

The hardest part was ... ☐ finding the missing shape ☐ using the row and the column ☐ checking the whole grid ☐ something else

What helped me: ☐ my tiles ☐ looking along and down ☐ checking my answer ☐ I was not stuck

Next time I will try: ☐ using my tiles ☐ checking every column ☐ checking my answer ☐ asking a question

☐ I am still stuck (my grown-up will photograph this page for my tutor)

☐ I would like to pass

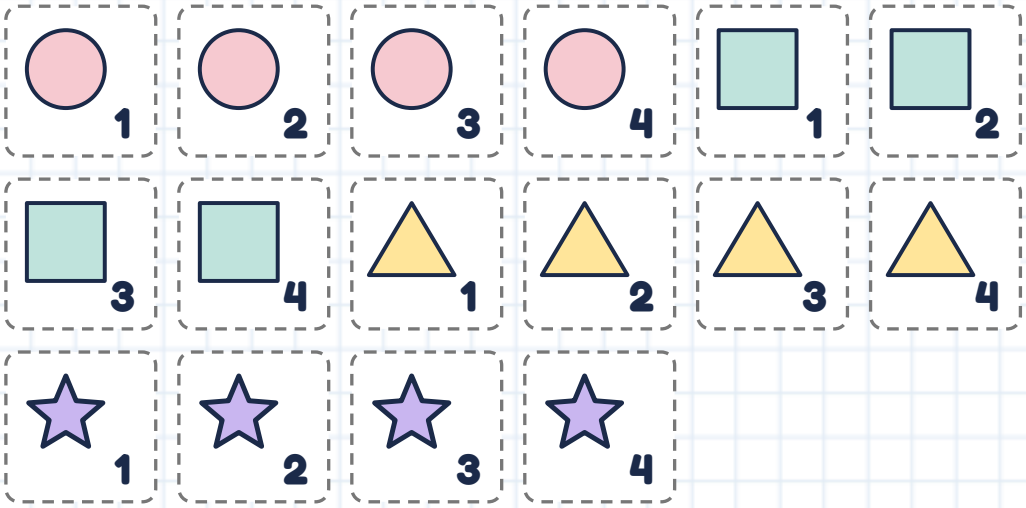
Name _____ Date _____

Kit page: grid tiles and mats

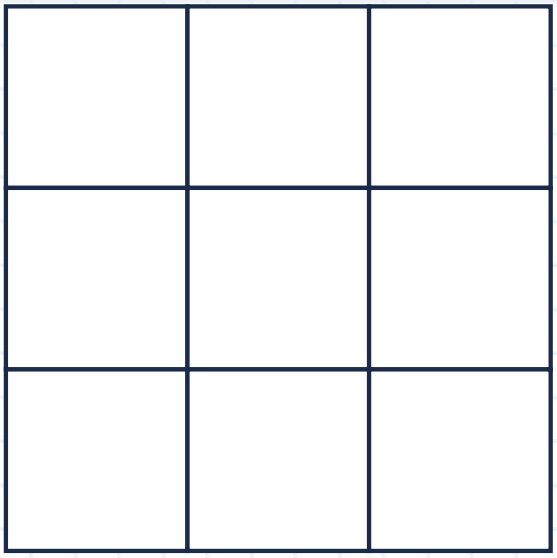
A grown-up cuts out all 16 tiles before the lesson: 4 circles, 4 squares, 4 triangles and 4 stars, each with a digit 1 to 4. The two mats are not cut. We use 9 tiles on mat 1 in class and all 16 tiles for Go further. No printer or no cutting? Use nine small things of three kinds on a 3 by 3 grid you draw. Keep the tiles in an envelope.

Print this page on its own sheet, or on one side only, so cutting never spoils another page. A grown-up cuts; children do not use scissors. Keep the pieces in an envelope with your name. We use them again.

Tiles: cut out all 16 (a grown-up cuts). The mats are not cut.



Mat 1 (3 by 3)



Mat 2 (4 by 4)

